SUMMER EXAMINATION 2026

Artificial Intelligence

BSIT - 3rd Semester Time: 90 Minutes Max Marks: 80

Name: ______________ Roll No.: ______________ Date: ______________

1. AI is best described as the study of agents that perceive and act
(A) intelligent behavior (B) database normalization
(C) file compression (D) packet framing

2. The Turing test evaluates whether a machine can
(A) converse indistinguishably from a human (B) sort numbers
(C) store facts (D) encrypt files

3. Weak AI is designed for
(A) a limited task (B) every intellectual task
(C) only robotics (D) only games

4. A rational agent chooses an action that maximizes
(A) expected performance (B) memory usage
(C) program length (D) randomness

5. In PEAS, the E stands for
(A) Environment (B) Execution
(C) Encoding (D) Evaluation

6. A fully observable environment provides the agent with
(A) complete relevant state information (B) only rewards
(C) no sensors (D) past actions only

7. A reflex agent selects actions mainly from
(A) the current percept (B) a learned value function
(C) a planning graph (D) a proof tree

8. A model-based agent maintains
(A) an internal state (B) only a lookup table
(C) a random seed (D) a utility invoice

9. A utility-based agent uses utility to
(A) rank desirable outcomes (B) tokenize text
(C) prune syntax trees (D) count database rows

10. An episodic task environment has actions that
(A) do not affect later episodes (B) always require planning
(C) must be continuous (D) have no observations

11. Breadth-first search expands the
(A) shallowest frontier node (B) lowest heuristic node
(C) deepest node (D) random node

12. Depth-first search uses a frontier organized as a
(A) stack (B) queue
(C) priority table (D) hash ring

13. Uniform-cost search selects the frontier path with
(A) lowest accumulated cost (B) lowest depth
(C) highest heuristic (D) most children

14. A heuristic function estimates
(A) cost from a state to a goal (B) the exact training label
(C) network bandwidth (D) number of agents

15. A star search evaluates a node with
(A) f(n)=g(n)+h(n) (B) f(n)=g(n)-h(n)
(C) f(n)=h(n)/g(n) (D) f(n)=g(n)h(n) only

16. A heuristic is admissible when it
(A) never overestimates true remaining cost (B) always equals true cost
(C) is negative (D) ignores the goal

17. If every edge costs 1, a path of depth 4 has cost
(A) 4 (B) 1
(C) 0 (D) 8

18. Hill climbing can fail at a
(A) local maximum (B) root node only
(C) terminal compiler (D) balanced database

19. Simulated annealing sometimes accepts
(A) a worse move (B) only a goal move
(C) no move (D) a duplicate state

20. Minimax assumes that the opponent
(A) acts optimally (B) acts randomly
(C) never moves (D) maximizes our score

21. Alpha-beta pruning changes minimax results
(A) never (B) only at even depth
(C) always (D) only with chance nodes

22. In alpha-beta pruning, alpha is the best value found for
(A) MAX so far (B) MIN so far
(C) the root heuristic (D) a random leaf

23. A constraint satisfaction problem consists of variables, domains, and
(A) constraints (B) rewards
(C) gradients (D) tokens

24. Backtracking search usually assigns
(A) one variable at a time (B) all variables randomly
(C) only goal variables (D) no values

25. The MRV heuristic chooses the variable with
(A) fewest legal values (B) largest domain
(C) highest cost (D) most neighbors only

26. Arc consistency removes values that
(A) have no supporting neighbor value (B) are globally optimal
(C) are duplicated (D) are outside the root

27. In STRIPS, an action has preconditions and
(A) effects (B) pixels
(C) gradients (D) priors

28. Forward planning starts at
(A) the initial state (B) the goal state
(C) the deepest leaf (D) the utility maximum

29. Backward planning works backward from
(A) the goal (B) the first percept
(C) the largest domain (D) the reward table

30. A proposition is a statement that is
(A) true or false (B) always numerical
(C) a probability distribution (D) a neural layer

31. Forward chaining applies rules from
(A) known facts toward conclusions (B) goals toward facts
(C) random facts (D) only contradictions

32. Backward chaining begins with
(A) a query goal (B) an empty database
(C) a sensor reading (D) a heuristic value

33. A Bayesian network represents dependencies using a
(A) directed acyclic graph (B) stack
(C) confusion matrix (D) parse tree only

34. Naive Bayes assumes features are conditionally independent given the
(A) class (B) learning rate
(C) action (D) cluster index

35. Fuzzy logic allows truth values
(A) between 0 and 1 (B) only -1 and 1
(C) only 0 or 1 (D) above 100

36. Supervised learning uses
(A) labeled examples (B) no observations
(C) only rewards (D) only rules

37. Unsupervised learning seeks structure in
(A) unlabeled data (B) test labels
(C) utility functions (D) action policies

38. Reinforcement learning learns from
(A) rewards and penalties (B) class labels only
(C) SQL joins (D) grammar rules

39. A validation set is used mainly for
(A) model selection and tuning (B) final unbiased reporting
(C) data collection (D) encryption

40. Overfitting means a model fits training data well but
(A) generalizes poorly (B) has no parameters
(C) cannot train (D) has low variance always

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1. High bias commonly causes
(A) underfitting (B) memorization
(C) data leakage (D) high variance only

2. Precision is the fraction of predicted positives that are
(A) actually positive (B) actually negative
(C) all observations (D) missed positives

3. Recall is the fraction of actual positives that are
(A) correctly found (B) predicted negative
(C) all negatives (D) true negatives only

4. For TP=8 and FP=2, precision is
(A) 0.80 (B) 0.20
(C) 0.60 (D) 0.90

5. The F1 score is the harmonic mean of
(A) precision and recall (B) accuracy and loss
(C) bias and variance (D) speed and memory

6. A perceptron computes a weighted sum followed by
(A) an activation (B) a database join
(C) a graph search (D) a token split

7. Gradient descent updates parameters using the
(A) negative loss gradient (B) positive labels only
(C) largest feature (D) random action

8. A convolutional neural network is especially suited to
(A) spatial patterns (B) symbolic proofs only
(C) transaction logs only (D) shortest paths

9. Pooling in a CNN commonly reduces
(A) spatial dimensions (B) the number of classes
(C) training labels (D) token vocabulary only

10. An RNN is designed to process
(A) sequences (B) static tables only
(C) graphs only (D) images without order

11. An LSTM uses gates to help preserve
(A) long-range dependencies (B) database keys
(C) pixel colors only (D) search depth

12. A Markov decision process includes states, actions, transition model, rewards, and
(A) a discount factor (B) a compiler
(C) a parse tree (D) a domain vocabulary

13. A policy maps states to
(A) actions or action probabilities (B) class labels only
(C) database rows (D) heuristic costs

14. Q-learning estimates
(A) action values (B) class priors only
(C) parse probabilities (D) pixel edges

15. The Q-learning update uses the reward plus discounted
(A) next-state best Q value (B) current input size
(C) training accuracy (D) heuristic depth

16. Exploration in reinforcement learning means
(A) trying uncertain actions (B) always choosing the best known action
(C) removing rewards (D) stopping training

17. A false positive occurs when a negative case is
(A) predicted positive (B) predicted negative
(C) left unlabeled (D) counted twice

18. Data leakage occurs when training uses information
(A) unavailable at prediction time (B) from a larger batch
(C) with missing values (D) after normalization

19. Responsible AI emphasizes fairness, transparency, privacy, and
(A) accountability (B) unlimited surveillance
(C) opaque decisions (D) forced automation

20. A model card is intended to document
(A) model purpose, limits, and performance (B) network routes
(C) database indexes (D) source code licenses only

21. Differential privacy protects individuals by adding controlled
(A) noise (B) labels
(C) search depth (D) network hops

22. A vector dot product measures alignment and is largest for vectors pointing
(A) in similar directions (B) in opposite directions
(C) at random (D) with zero length only

23. The determinant of a 2 by 2 matrix [[a,b],[c,d]] is
(A) ad-bc (B) ab+cd
(C) ac-bd (D) a+b+c+d

24. If a fair binary event has probability 0.5, its odds of success are
(A) 1 to 1 (B) 1 to 2
(C) 2 to 1 (D) 0 to 1

25. Cross-entropy penalizes a confident prediction that is
(A) wrong (B) correct
(C) uniform (D) missing a feature

26. A confusion matrix summarizes
(A) classification outcomes (B) regression slopes
(C) search paths (D) cluster centroids

27. A ROC curve plots true positive rate against
(A) false positive rate (B) true negative rate
(C) loss (D) sample size

28. K-means assigns points to the nearest
(A) centroid (B) decision stump
(C) support vector (D) parse node

29. PCA projects data onto directions of greatest
(A) variance (B) class count
(C) reward (D) text frequency only

30. The principal purpose of normalization in ML is to
(A) put features on comparable scales (B) create labels
(C) increase leakage (D) remove all outliers

31. A learning rate that is too large may cause training to
(A) diverge (B) stop overfitting automatically
(C) produce labels (D) become deterministic

32. A learning rate that is too small usually makes convergence
(A) slow (B) impossible in every case
(C) random (D) independent of data

33. An agent's sensors provide percepts, while its actuators produce
(A) actions (B) heuristics
(C) constraints (D) utilities

34. Iterative deepening combines the low memory of DFS with the completeness of
(A) BFS (B) hill climbing
(C) minimax (D) beam search

35. A goal test checks whether a state satisfies
(A) the goal condition (B) the heuristic estimate
(C) the action cost (D) the percept history

36. Knowledge engineering converts domain expertise into
(A) a knowledge base (B) a pixel matrix
(C) a routing table (D) a loss curve

37. The utility of an outcome represents its
(A) desirability (B) token count
(C) search depth (D) probability only

38. An expert system usually contains a knowledge base and an
(A) inference engine (B) image encoder
(C) order queue (D) optimizer

39. The discount factor gamma is normally constrained to the interval
(A) 0 to 1 (B) 1 to 2
(C) -2 to -1 (D) 10 to 100

40. An AI agent's performance measure specifies
(A) how success is evaluated (B) how files are compressed
(C) how tokens are split (D) how keys are stored